

SUMMARY

Climate risk and resilience specialist with 8 years of experience across county and city governments, clean energy companies and national planning programs in the United States, India and Bangladesh. I take projects from raw data to a funded plan: flood and extreme heat risk assessments, GIS and dashboard tools that officials and residents use, carbon accounting and net-zero strategy, and the grants and briefings that turn analysis into action. Registered architect and environmental planner; MS in Sustainability Management, Columbia University.

SKILLS

Climate risk & resilience

Vulnerability assessment · flood, sea level rise & storm surge · extreme heat & health · scenario analysis · Monte Carlo

GIS, data & dashboards

ArcGIS Pro · ModelBuilder · QGIS · Experience Builder · StoryMaps · Survey123 · Power BI · Excel

Carbon & sustainability

GHG Protocol Scope 1–3 · LCA (openLCA) · MACC · net-zero roadmaps · GRI · climate finance · grants

EXPERIENCE

Broward County Government

2025 – Present

Natural Resource Senior Specialist (Climate Resilience & Sustainability) · Fort Lauderdale, FL

- **Countywide Extreme Heat Initiative** **NACO 2025 AWARD** – Started and ran the county's heat program for ~2 million residents, in-house with no added budget. Ranked heat risk from satellite surface temperature, green cover and social vulnerability data; linked 3,300+ heat-related ER visits to the daily heat index; worked with 24 cities to set up 39 cooling centres; coordinated heat alerts with the National Weather Service; briefed town halls, doctors and community groups.
- **Heat tools** – Built the public Extreme Heat StoryMap and an Urban Heat Dashboard showing 327 schools, libraries and clinics by heat level, with a heat index forecast and cooler walking routes. Built a Power BI heat-illness dashboard (public release pending data approval).
- **Vulnerability Assessment** – Built automated ArcGIS ModelBuilder workflows to measure flood depth and sensitivity for 700+ county assets under 30 future scenarios, from MIKE SHE model outputs; led QA/QC and documentation to Florida DEP standards.
- **Resilience Plan Project Tracker** – Designed and launched a public GIS dashboard of 45,902 proposed resilience projects worth ~\$28 billion, filterable by city, district, type, phase and funding.
- **Resilient Broward Grant Program (\$20M)** – Managed the program day to day; coordinated with every city and water control district, ran applicant presentations and webinars, built the program website, ArcGIS application portal and project dashboard.
- **Site and emergency flood analysis** – Modelled 100-year rainfall, storm surge and sea level rise for a planned forensic science building (an evacuation centre) in 3D, which changed its site design; mapped safe evacuation routes to shelters using a two-foot flood depth limit.
- **Energy and emissions** – Built public dashboards for a 199 kW county solar array and county EV charging (with demand forecasting); estimated employee commuting emissions for the Scope 3 inventory with EV adoption scenarios; redesigned the resilience plan's economic benefits and adaptation cost dashboards in Power BI.

SOLARCYCLE

2025

Sustainability Consultant (contract) · Solar panel recycling

- Ran cradle-to-grave LCAs for several solar panel types; built the corporate GHG inventory and comparative LCAs to improve the recycling process.
- Built a marginal abatement cost curve and pre-feasibility checks to rank cuts by CO₂e saved, cost and return; wrote the company-wide net-zero strategy and roadmap.
- Co-authored the white paper *Reuse vs. Recycling: A Life Cycle Assessment of Solar Panel End-of-Life Pathways* (2026).

City of Charlotte, North Carolina

2024

EDF Climate Corps Fellow · Environmental Defense Fund

- Ran a city-wide climate risk assessment: screened heat, flood and other hazards, assessed exposure of assets, people and services, and built risk matrices to rank adaptation priorities.
- Led the spatial analysis for a ~\$1M EV charging (EVCI) grant application, choosing sites by demand gaps, equity and access, with funder-ready maps.

IPE Global

2021 – 2023

Senior Analyst, Environmental Planning · Gurgaon, India

- **National Tourism Master Plan, Bangladesh** (USD 2M assignment, ~USD 1.08B investment pipeline; adopted by government) – Ran GIS climate and disaster risk assessments, mapped sensitive areas and set land-use controls; drafted environmental and social safeguards; supported national consultations.
- **Lakshadweep Island Master Plans** (USD 90M, for NITI Aayog) – Land suitability and carrying capacity studies for five islands and lagoons; wrote the development control regulations and eco-tourism guidelines; supported DPRs for nine PPP projects.
- **Kavaratti Smart City** (USD 65M) – On-site project coordinator; led the solid waste assessment for 10 islands through to feasibility, cost estimates and tender; supported Blue Flag certification for four beaches.
- Supported proposals to the World Bank, ADB, IFC and the European Union, and built the firm's climate change portfolio.

National Institute of Urban Affairs (with GIZ)

2020 – 2021

Subject Matter Expert · Climate Smart Cities program · New Delhi

- Wrote two national training manuals (city climate action planning; urban biodiversity) and trained officials from 50+ city governments, including GIS exercises on climate risk mapping.

School of Planning and Architecture, New Delhi

2018 – 2021

Research Associate, Design Innovation Centre and Dept. of Environmental Planning · also Visiting Faculty (2021)

- **Urban Biodiversity Toolkit** – Co-developed a GIS-based toolkit and app for city biodiversity planning across four cities; granted Indian Patent No. 602097 (2026, co-inventor); presented for India at UNCCD COP 14 (2019); IISF Best Biodiversity Innovation Award (2020).
- **Air carrying capacity of Delhi** – Built emission inventories, mapped sources around 37 monitoring stations and set a carrying capacity framework and action plan for the city's pollution regulator.
- Rapid environmental assessment for a National Green Tribunal case; 100-year vision and guidelines for the 10,400 sq km Taj Trapezium Zone; planning framework for India's hilly regions for NITI Aayog.

EDUCATION

MS, Sustainability Management · Columbia University, New York

2024

Built a climate risk and scenario analysis course with Monte Carlo exercises. Client projects: Scope 1–3 strategy (Preqin), LCA for a 45Q carbon capture case (CarbonQuest), GHG inventory, GRI report, climate risk research (CalPERS), climate finance research (Power for All). Exchange: University of St. Gallen, Thrive program.

Master of Planning, Environmental Planning

2018

School of Planning and Architecture, New Delhi · First class with distinction

Bachelor of Architecture

2016

DCRUST, Sonapat, India · First class with distinction

AWARDS & CREDENTIALS

- William K. Bowes, Jr. Award for Leadership, EDF Climate Corps (2025)
- NACo Achievement Award, Countywide Extreme Heat Initiative (2025)
- Indian Patent No. 602097, biodiversity conservation system (2026)
- Best Biodiversity Innovation Award, India International Science Festival (2020)
- Registered Architect, Council of Architecture, India (2018)
- LEED Green Associate (2023)
- Associate Member, Institute of Town Planners, India

PUBLICATIONS

- 5 book chapters on air quality, urban biodiversity and regional planning – CRC Press (Taylor & Francis), IGI Global, Nova Science (2021–22)
- 2 research articles on climate education – Columbia University (2023, 2025)
- White paper on solar panel end-of-life LCA – SOLARCYCLE (2026)
- Full list at my portfolio website